

PRNOB KUMAR BARMAN

Baltimore, MD | 4439315511 | pbarman1@umbc.edu

PROFESSIONAL SUMMARY

AI & Machine Learning Engineer and **PhD candidate** with extensive research experience and **7+ years of hands-on application** in designing, training, and deploying intelligent systems across healthcare, cybersecurity, and financial services. Expertise in **LLMs, generative AI, recommendation systems, reinforcement learning**, and cloud-native MLOps. Proficient in **Agile/Scrum project management for ML initiatives** on AWS/GCP. Proven track record delivering production-grade ML services with HIPAA/SOC 2/GDPR compliance. Published researcher in fairness-aware AI and Explainable AI (XAI) at MedInfo, SIAM SDM, and AMIA.

CORE SKILLS

AI/ML: LLMs (GPT, LLaMA, Falcon), Generative AI, Transformers, RAG, Deep Learning, NLP, Recommendation Systems, Reinforcement Learning (PPO), Anomaly Detection, Explainable AI (XAI) | **MLOps & DevOps:** AWS (SageMaker, EKS, Lambda), GCP, Docker, Kubernetes, MLflow, Airflow, CI/CD, Workflow Management, Orchestration, SDLC | **Programming:** Python, SQL, R, PyTorch, TensorFlow, Hugging Face, LangChain, FastAPI, OOP, Microservices Architecture | **Data Engineering:** ETL, Kafka, Pinecone, FAISS, PostgreSQL, Redshift, Data Modeling, Data Warehousing | **Security & Compliance:** SIEM (Splunk, Wazuh), HIPAA, GDPR, NIST 800-53, SOC 2, Threat Intelligence | **Ethical AI:** SHAP, LIME, Fairlearn, Data Privacy, PII Protection, Algorithmic Bias Mitigation | **Testing:** Model Validation, Unit Testing, Integration Testing, Performance Testing

PROFESSIONAL EXPERIENCE

Machine Learning Engineer (PhD Intern) – Behavioral Health AI Platform **Nov 2025 – Present**
Technuf LLC *Rockville, MD*

- Managed end-to-end development of HIPAA and GDPR-compliant AI workflows on AWS SageMaker using Agile methodologies, coordinating with cross-functional teams to deliver preprocessing, training, and deployment pipelines for transformer-based models
- Led development of cloud-native MLOps platform integrating SageMaker, MLflow, Docker, and Kubernetes, facilitating sprint planning and daily stand-ups to orchestrate training workflows and automate staging-to-production promotion
- Developed **reinforcement learning recommendation engine** using PPO with rigorous model validation and performance testing protocols, achieving **15% increase in session retention** and 10% improvement in engagement
- Created comprehensive fairness and ethical AI toolkit using SHAP, LIME, and Fairlearn for evaluating demographic parity and algorithmic bias mitigation, guiding equitable model iterations
- Engineered real-time inference gateway with FastAPI microservices on EKS backed by Pinecone, delivering personalized recommendations with **sub-350ms latency**
- Built FHIR-compliant ETL pipeline using Kafka, Airflow, and BERT tokenization to process **1M+ behavioral health notes daily** while ensuring data privacy through anonymization techniques

Machine Learning Engineer (PhD Intern) – Embedded Cybersecurity **Jul 2025 – Nov 2025**
Axiado Corporation *San Jose, CA*

- Developed end-to-end ML pipelines for ARM-based embedded security processors, deploying optimized TensorFlow Lite and PyTorch models, resulting in **15% reduction in processing time** for anomaly detection on edge devices
- Implemented quantization, pruning, and compression techniques for edge hardware optimization with comprehensive unit and integration testing, achieving lower latency without sacrificing detection accuracy
- Designed behavior-based intrusion detection using temporal patterns from firmware logs with rigorous model validation, improving **early-stage threat detection accuracy by 20%** compared to rule-based baselines
- Built automated SDLC workflows for embedded deployment including ONNX export, hardware-specific optimization, and regression testing to ensure consistent model performance

Research Assistant – Recommendation Systems & Fairness-Aware AI **Aug 2023 – Present**
University of Maryland, Baltimore County (UMBC) *Baltimore, MD*

- Led research team of 2 graduate students on hybrid recommendation systems combining DistilBERT embeddings and collaborative filtering for personalized healthcare support group formation
- Built neural collaborative filtering architectures with comprehensive model validation and performance testing, achieving **25% gain in NDCG@10 and 15% gain in hit-rate@10** compared to traditional methods
- Designed cold-start resolution module using contrastive learning and multi-task objectives, improving onboarding efficiency by **30%**
- Mentored 3 undergraduate researchers on advanced NLP techniques and ethical AI practices, resulting in **20% improvement in model fairness metrics**
- Authored and co-authored **peer-reviewed publications** at MedInfo, SIAM SDM, and AMIA on healthcare AI and fairness-aware recommendation systems

Machine Learning Engineer (Graduate Intern) – Generative AI & Prompt Engineering **Jun 2024 – Aug 2024**
Technuf LLC *Rockville, MD*

- Managed ML project using Agile/Scrum methodologies, facilitating sprint retrospectives to design RAG pipelines using LangChain, ChromaDB, and FAISS for enhanced threat intelligence and security operations
- Built hybrid threat scoring engine combining LightGBM and LSTM models with comprehensive integration testing, reducing **low-value alerts by 30%** and improving analyst efficiency by **25%**
- Created structured prompts for GPT-style models to classify alerts and summarize threat activity, integrating outputs into Power BI dashboards for SOC teams with **40% faster incident response**
- Developed cybersecurity-specific NER pipelines using spaCy and fine-tuned transformers with model validation protocols to extract IOCs in near real-time

Machine Learning Engineer (Graduate Intern) – Conversational AI

Aug 2022 – Dec 2022

Technuf LLC

Baltimore, MD

- Architected multi-turn conversational AI system using BERT embeddings and GRU-based dialog management for healthcare scheduling, improving patient engagement by **18%**
- Implemented intent classification and entity extraction models with rigorous unit and integration testing, achieving **95% accuracy** across patient queries
- Built PHI redaction pipeline ensuring HIPAA and GDPR compliance through anonymization techniques, combining regex rules and BERT-based NER ensembles
- Developed escalation risk classifier using XGBoost with model validation protocols, **reducing response times by 22%** through improved agent routing

Deputy Director – AI Systems & Policy Modeling

Apr 2018 – Dec 2021

Bangladesh Bank

Dhaka, Bangladesh

- Led AI modernization initiatives managing a team of **12 data scientists and engineers** across policy analytics, AML monitoring, and digital customer interaction projects
- Directed development of Bengali-language NLP chatbot using Agile methodologies, mentoring 5 junior engineers and **reducing call center volume by 25%**
- Designed anomaly detection pipelines using autoencoders and isolation forests with comprehensive model validation on large-scale transactional data, identifying **200+ potential AML cases annually**
- Built policy forecasting models with system design best practices including ARIMA, VAR, and scenario analysis tools, integrating ETL workflows into Tableau dashboards for senior leadership

EDUCATION

PhD in Information Systems (Human-Centered Generative AI, Fairness-Aware ML, & Explainable AI) Expected May 2026

University of Maryland, Baltimore County (UMBC) – Baltimore, MD

Master of Science in Statistics

May 2021

University of Kentucky – Lexington, KY

Master of Science in Statistics

Jan 2017

University of Dhaka – Dhaka, Bangladesh

Bachelor of Science in Statistics

Sep 2015

University of Dhaka – Dhaka, Bangladesh

SELECTED PUBLICATIONS

- **Barman, P. K.**, Reynolds, T. L., & Foulds, J. (2025). Facilitating online healthcare support group formation using topic modeling. *Studies in Health Technology and Informatics*, 329, 1049–1053. <https://doi.org/10.3233/SHTI250999>
- Balogun, Z. A., **Barman, P. K.**, Onwumbiko, B. K., & Reynolds, T. L. (2025). User experiences with mobile personal health records. *AMIA Annual Symposium Proceedings, 2024*, 162–171.
- **Barman, P. K.**, Reynolds, T. L., & Foulds, J. (2025). *Enhancing online support group formation using topic modeling* [Poster]. SIAM International Conference on Data Mining.
- Barman, P. K. (2024). *Detecting abnormal health conditions in smart home using a drone* (arXiv:2310.05012). <https://arxiv.org/abs/2310.05012>

AWARDS AND GRANTS

NSF Travel Grant – SIAM SDM Doctoral Forum Presenter, 2025

UMBC Graduate Student Association Research Grant – Healthcare Recommender Systems, 2024